

Question 1:

1. tf for message 2 : $[0, \frac{1}{5}, 0, 0, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, 0]$

2. IDF for message 2:

$$[\log(\frac{3}{1}), \log(\frac{3}{3}), \log(\frac{3}{2}), \log(\frac{3}{2}), \log(\frac{3}{1}), \log(\frac{3}{1}), \log(\frac{3}{1}), \log(\frac{3}{1}), \log(\frac{3}{1})]$$

3. $TF-IDF = TF \times IDF$

$$= [0, 0, 0, 0, 0.219, 0.219, 0.219, 0.219, 0]$$

4. Yes, since "are good" is appear in all the messages. So it can not help us to distinguish those messages. That why IDF of "are good" is $\log(\frac{3}{3}) = 0$.

5. "are" and "good" will be 0. Since they all exist in all 3 messages. So IDF of "are" & "good" will be $\log(\frac{3}{3})$ which is zero. Since "are" & "good" can not help us distinguish those messages.

Question 2:

Support means the real number of samples in datasets for specific class.

$$\text{Support} = TP + FN$$

Question 3:

No, accuracy not same with F1-score.

If dataset highly unbalanced, accuracy always misleading.

example: 100 samples $\left\{ \begin{array}{l} 99 \text{ positive} \\ 1 \text{ negative} \end{array} \right.$ classifier predict all positive.

	0	1	
0	99	0	
1	1	0	

so Accuracy will be 99% but F1-score = 0

Thus, F1-score is better for unbalanced data.

Question 4:

We can give more weight for bad data. Which means assign higher weights to minority class.